

Annex 2

WHO points to consider in continuous manufacturing of pharmaceutical products

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Contents

1. Introduction	96
2. Glossary	97
3. Benefits of and challenges in continuous manufacturing	98
3.1 Technical and operational challenges	99
3.2 Regulatory challenges	100
3.3 Workforce challenges	100
4. Good practice considerations	100
5. Risk management	101
6. Control strategy	102
7. Process dynamics	104
8. Computerized systems	105
9. Validation and verification	105
10. Stability testing	105
References	106
Further reading	106



1. Introduction

Manufacturers of active pharmaceutical ingredients (APIs), excipients and finished pharmaceutical products (FPPs) have predominantly used batch processing as a means to manufacture their products.

In recent years, several manufacturers have opted to introduce continuous manufacturing (CM) in pharmaceutical production. Although CM is a relatively new approach in pharmaceutical product manufacturing, this concept has been used in other industries for nearly a century. There are several examples of industries utilizing this approach, including oil refining, metal smelting, petrochemical product manufacturing, and certain excipient, food and beverage manufacturing.

CM (including flow chemistry, where appropriate) can be applied to the production of certain chemicals, starting materials (excipients and APIs) and FPPs.

CM requires automated systems, such as metering of input materials (at a minimum), to ensure a sustainable and controlled manufacturing process.

In a CM process, the input materials are continuously fed into and transformed within the process, and the processed output materials are continuously removed from the system. This description can be applied to an individual unit operation or the entire manufacturing process consisting of a series of unit operations.

CM does not have to be end-to-end in production of a product. It could be applied to some (semicontinuous) or all unit operations in a manufacturing process. As an example, in the production of oral solid dosage forms, some steps (such as feeding and mixing) or all steps can be included in CM.

Although the amount of material being processed at any given instance may be relatively small in a CM process, the process can run over a period of time to generate desired quantities of finished material meeting the necessary quality standard.

In the traditional batch manufacturing process, sampling and testing of samples after certain processing steps are the norm. This often leads to down times and hold times. The relatively new approach of CM, utilizing sensors for in-line and on-line analytical testing, may reduce such down and hold times. It may further facilitate utilizing the full capacity of equipment and production lines, reduce human error, and support quality control and testing.

This document presents points to consider for manufacturers implementing CM in the production of APIs and FPPs. Although the examples given in the document focus on oral solid dosage forms, the principles may be applied to other dosage forms, biologicals (such as therapeutic proteins) and vaccines (for example, those manufactured by novel technology). This document complements the International Council for Harmonisation of Technical

Requirements for Pharmaceuticals for Human Use (ICH) Q13 harmonised guideline *Continuous manufacturing of drug substances and drug products* ([1](#)) and should be read alongside ICH Q13 and WHO good manufacturing practices.

2. Glossary

at-line. A system whereby samples are collected manually and the analyser is located next to the process.

batch. A specific quantity of material produced in a process or series of processes so that it is expected to be homogeneous within specified limits. In the case of continuous production, a batch may correspond to a defined fraction of the production. The batch size can be defined either by a fixed quantity or by the amount produced in a fixed time interval.

batch size. The specified quantity of input material; or the quantity of output material or product; or the time it takes to manufacture the specified quantity of output material or product; or a combination of the above.

continuous manufacturing. Continuous feeding of input materials into a process with processed output materials continuously removed from the system, consisting of two or more integrated unit operations.

flow chemistry. Flow chemistry is also known as continuous flow or plug flow chemistry; it involves a chemical reaction run in a continuous flow stream rather than a batch production.

in-line. A process analytical technology system that is incorporated into the flow of the process and produces continuous data without sampling using capacitance, light scattering, spectroscopy, on-line liquid chromatography and other types of sensors.

on-line. A system that is connected directly to the process and collects and automatically analyses samples, which are never returned to the process.

pharmaceutical product. Any material or product intended for human or veterinary use presented in its finished dosage form or as a starting material for use in such a dosage form, that is subject to control by pharmaceutical legislation in the exporting or importing state.

process analytical technology. A mechanism to design, analyse and control manufacturing processes through the measurement of quality attributes of intermediates or products and monitoring of process parameters that affect the critical quality attributes.

process control. The practice of monitoring and adjusting a process to achieve a desired outcome. It is a combination of engineering and statistics that involves using algorithms, mechanisms and architectures to maintain the output of a process within a specific range.

process dynamics. The response of a manufacturing process to changing inputs, conditions or transient events. A transient event is a temporary condition in which a process goes through a dynamic change. This change may be due to a disturbance or an intentional alteration in the selected operating conditions (for example, start-up, shutdown or changes from one operating condition to another).

quality by design. A systematic approach to development that begins with predefined objectives and emphasizes product and process understanding and process control, based on sound science and quality risk management.

real-time release testing. The ability to evaluate and ensure the quality of in-process or final product based on process data, which typically include a valid combination of measured material attributes and process controls.

residence time distribution. A measure of the range of residence times experienced by material passing through a specific process environment, vessel or unit operation.

state of control. A condition in which the set of controls consistently provides assurance of continued process performance and product quality.

For other definitions, see the WHO Quality Assurance of Medicines Terminology Database: <https://www.who.int/publications/m/item/quality-assurance-of-medicines-terminology-database>.

3. Benefits of and challenges in continuous manufacturing

The benefits and challenges of CM have been described in various guidelines and articles (see “further reading” section).

CM may result in increased output of product in a shorter time frame than traditional batch processing. It may also provide safety benefits due to lower exposure risk to operators.

CM may, in certain cases, support economic and environmental sustainability. Consideration should be given to environmental impact, water and energy consumption, and evaluation of capital and operating costs.

CM may also present challenges to manufacturers. These include providing specialized training of personnel in the new concept, managing product changeover and cleaning of such lines.

With the technical, operational and economic challenges, as well as the risks associated with CM, it is important that manufacturers wanting to move from batch manufacturing to CM processes ensure that there is sufficient process knowledge to facilitate risk management and the development and implementation of an appropriate control strategy. Ingredient and process knowledge, through education and characterization, is needed to ensure an appropriate control strategy if transitioning from batch to continuous operations.

Challenges to adopting CM in the pharmaceutical industry include:

- technological issues (for example, process knowledge);
- logistical concerns (for example, new equipment and computerized systems);
- advanced control strategies comprising complex analytical instrumentation and technology for improved process control using robust and reliable methods (for example, in-line, on-line and at-line analytics);
- real-time data strategies for critical quality attributes and critical process parameters (for example, to maintain a state of control);
- regulatory uncertainty (for example, different regulatory requirements in countries);
- integrating downstream unit operations (for example, semicontinuous manufacturing);
- personnel (for example, specific training and qualification);
- risks (including actual and perceived risks);
- economic issues (for example, return on investment);
- flexibility issues (for example, adjusting process).

Batch-to-batch variability and purity for clinical applications is still one of the biggest challenges with implementation (that is, continuous processing to meet current good manufacturing practices and good laboratory practices).

3.1 Technical and operational challenges

The lack of commercially available equipment suitable for small-scale CM lines presents a challenge to formulation and development facilities as well as commercial manufacturers.

The importance of the link between batches produced and used during clinical trials and commercial batches should also not be underestimated.

Operators should have knowledge of the complexities relating to process control and associated risks due to equipment, sensor or computerized system failure, leading to a lack of continuous material flow, overfilling,

overpressurization, spills and backflow of material. Hence, operators should be well trained and qualified.

3.2 Regulatory challenges

Furthermore, while CM aligns strongly with international guidelines, such as those of the United States Food and Drug Administration and the ICH, pharmaceutical manufacturing is a global enterprise, and companies must gain approval for their products in multiple countries with their own regulatory bodies. Not all regulatory agencies may have established requirements and standards for CM.

When changing from an existing batch manufacturing process to a CM process, the continuous process can be introduced as a new process for a new product or as a post-approval manufacturing change. Where required, comparable quality and bioequivalence should be demonstrated (see also ICH Q13 ([1](#)) risks, change and training).

More work and harmonization may be required to resolve issues related to regulatory challenges and requirements of CM.

3.3 Workforce challenges

Designing, implementing and adequately regulating new approaches in manufacturing require personnel who are skilled and well trained in manufacturing as well as in the regulatory environment.

Additional training may be required for personnel in production, quality control, quality assurance, engineering and the regulatory function, as specialized equipment, sensors, feedback systems, computerized systems and data management may be required in CM.

4. Good practice considerations

Good practice considerations in CM should start at the product and process development stage.

CM may require manufacturers to acquire new equipment; implement new ways of managing existing equipment; acquire new instruments, computerized systems and software; and establish new modes of operation. This may further include a move to apply process analytical technology and principles of quality by design.

The premises and equipment should be appropriate to support CM. Equipment should remain within operating specifications over the duration of the CM process.

The pharmaceutical quality system should be suitably designed and appropriately implemented and maintained ([2](#)), including through material and

process management; qualification and validation; and maintenance, calibration and monitoring. Other pharmaceutical quality system elements include risk management, process capability index, process performance index, managing deviations, managing incidents and non-conformities (including material diversion and disturbance), product stability, and a control strategy.

Raw and starting materials used in a product should be traceable and consistently meet predefined specifications. As multiple lots of raw materials may be introduced at various times in CM, traceability is essential. This will require understanding of residence time distribution and process dynamics.

Systems should be in place for product collection, control and diversion of non-conforming material, and product rejections. Understanding of residence time distribution and process dynamics is required.

Other systems to be considered include:

- detailed start-up and shutdown procedures;
- how production collection and in-process sampling will occur as a means of assuring continued process performance and product quality;
- process validation and continued process verification procedures;
- personnel training procedures;
- cleaning and cleaning validation.

Appropriate resources should be provided.

Process models can be used for development of a CM process or optimization of process parameters to establish state of control, diversion strategy, material traceability and process dynamics.

Process parameters (including their settings and real-time data) should be controlled and monitored as part of the control strategy.

5. Risk management

Risk identification, risk and harm assessment, risk control and risk communication should be integral parts of the pharmaceutical quality system where CM is employed ([3](#), [4](#)).

Quantitative or qualitative analysis using an appropriate tool – for example, failure mode and effects analysis or a risk matrix – may be considered when doing risk assessment.

Risk assessment should be done at various stages in the life cycle of a product, from development through transfer of technology to commercial CM.

Risks relating to the sourcing and control of material, equipment, processing steps and cleaning, as a minimum, need to be included in the assessment. Each processing step or unit operation should be mapped out

indicating, as appropriate, quality attributes and process parameters. Risks and harms can then be assessed and controls identified.

Consideration should be given to:

- input material attributes, including their impact on process operations and product quality (note: process performance may vary where input material attributes are not consistent – flowability, particle size and distribution, cohesion, hygroscopicity and other attributes should be considered in the selection of material, and the manufacturer of the materials should be qualified as part of vendor qualification);
- process steps, including operating parameter settings (for example, time, temperature, rotation per minute, amperage, speed, and pressure during sifting, milling, blending, granulation, drying, compression, filling, sealing and coating);
- unexpected disturbances, possible deviations and non-conformances (for example, poor material flow, vibration, product build-up and material diversion).

The performance of computerized systems and the risks and impact associated with failure of such systems should be considered.

Appropriate means should be identified for the detection and handling of non-conforming material.

Risk assessment should be thorough to provide assurance that the required controls are identified and are effective to ensure that a state of control is achieved.

6. Control strategy

The control strategy for CM in commercial production should be initiated during the developmental phase of pharmaceutical products.

The control strategy should be based on the outcome of the risk assessment.

The control strategy should be clearly defined and describe all steps to ensure that the state of control is achieved. Relevant elements include input materials, residence time distribution, process dynamics, process monitoring, material diversion, real-time release testing (if applicable), specifications and process equipment.

The control strategy should have the ability to detect process departures, thereby enabling timely corrective actions to be taken to bring the process back into conformance.

Achieving and maintaining a state of control require appropriate measures to be taken relating to the management of raw and starting materials;

specifications; traceability; process monitoring; sampling; intermediates; equipment; and product collection and rejection.

Mechanisms should be in place to identify any drift in parameters or data trend that may be of concern. The root cause should be identified for significant transient events to ensure that appropriate action is taken.

In addition to traceability, it may be necessary to have additional controls when multiple lots of a raw material are used during CM.

Maintaining a state of control should provide assurance of consistent and desired product quality.

In ensuring that the manufacturing process is in a state of control, at least the following aspects should be considered and clearly defined:

- start-up, pauses and shutdown;
- in-process monitoring and control, with material collection and diversion of non-conforming materials;
- critical process parameters and critical quality attributes at various stages in the process.

Flowcharts indicating processing steps, continuous and semicontinuous steps and clear indication of location of sensors and probes may be useful.

The appropriate means of monitoring the process should be implemented. Sampling should be defined. This includes a clear description of the number of samples; frequency of sampling; sample size; sample location; in-line, on-line, off-line or at-line sampling; and limits and acceptance criteria.

The objectives of sampling, collecting and processing data should be clear, as data may be used in statistical analysis and trending. Set points and control limits should be appropriate. It may be possible to apply new approaches in technology and methodology in CM. CM affords more flexible handling compared to traditional batch manufacturing. Process parameters may be adjusted during processing based on measuring and results of quality attributes of the intermediate or in-process material, in real time, using (for example) process analytical technology. In-line sensors and devices may be useful to enable real-time identification of departures from expected results.

Validated systems should be in place to manage diversion of potentially non-conforming materials.

Based on process knowledge and understanding, the elements of a control strategy for CM include:

- measuring process parameters and critical quality attributes in a timely manner;
- maintaining the process in a state of control;
- maintaining the product attributes within specifications;

- appropriate use of artificial intelligence;
- optimizing process operation;
- realization process operation;
- realizing process efficiency improvements.

7. Process dynamics

Scientific knowledge supported by experimental data of the process dynamics and variables are needed to ensure that CM processes are appropriately designed, managed and operated within a state of control. Knowledge is required of the differences that may exist between developmental batch and commercial batch processing.

Processing steps in CM need to be well controlled to ensure the production of uniform products and thus require different approaches in process parameter control and monitoring. As an example, fluctuation in feed of raw material may impact the quality of a blend. For improved monitoring and control of the processing parameters, manufacturers may have to consider specifically designed equipment and instrumentation, computerized systems, and feedback systems. The selection of equipment and instrumentation should be suitable for its intended purpose and process for semicontinuous and continuous manufacturing, as applicable.

To obtain meaningful results, consideration should be given to:

- instrument selection and suitability for process conditions;
- analytical procedures and their development to ensure accuracy and robustness;
- sensor and probe placement for representative data collection;
- critical process parameters (such as flow rate, particle size distribution, compression force), including maintenance, servicing and calibration;
- system performance (including accuracy, operating range and sensitivity);
- data integrity and compliance with good practice requirements, including computerized system validation;
- sampling strategy (method, location, frequency, representativeness and sample size);
- consistency in analysis across the expected concentration range;
- acceptance criteria aligned with product quality attributes and regulatory expectations.

8. Computerized systems

Computerized systems should be appropriate for their intended use.

Computerized systems should be appropriately validated and be able to ensure the integrity of data (5, 6).

9. Validation and verification

The principles of process validation, as described in WHO guidelines, should be considered (7). In addition, specific attention should be given to start-up and shutdown of the process, process run time evaluation, and the ability to detect process excursions. Examples may include process robustness, process flow rate and residence time.

In CM, careful consideration should be given to the manner in which process performance and quality attributes are consistently controlled by the control strategy.

Process monitoring is dependent on residence time distribution. Real-time data collection and analysis during sampling should be representative of the process to ensure that transients outside the state of control are detected.

Where a traditional approach in process validation is followed, consideration should be given to the number of batches required for process validation. Any variation in results of attributes between different batches should be within an acceptable range. Consideration should be given to the possible impact on the process capability where differences in quantities and times (resulting in different batch sizes) in different batches are employed in CM.

10. Stability testing

Stability data for products manufactured by means of CM should be available. The same principles for stability testing apply as outlined in WHO guidelines for stability testing (8).

The selection of batches and the number of batches of product that should be subjected to stability testing should be justified.

Consideration should be given to variables that may be impacting the batches, such as the number of batches of input material to the batch of product and batch size.

Unless otherwise justified, stability data from commercial batches should be available and derived from batches where the state of control has been demonstrated.

Consideration should be given to the impact on stability of changes in such elements as run time, scale-out and process flow rate.

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